

CF09 CMAN

Kay Hynes

1) Dominant kernel

The dominant kernel is the 3×3 INT8 convolution layer from YOLO-nano, implemented as im2col + GEMM.

Given:

$$H = W = 52, \quad C_{in} = 64, \quad C_{out} = 128, \quad K_h = K_w = 3$$

$$H_{out} = W_{out} = 52$$

Equivalent GEMM:

$$M = H_{out} W_{out} = 52 \cdot 52 = 2704$$

$$K = K_h K_w C_{in} = 3 \cdot 3 \cdot 64 = 576$$

$$N = C_{out} = 128$$

So the kernel is:

$$C_{2704 \times 128} = A_{2704 \times 576} B_{576 \times 128}$$

Activations and weights are signed INT8. Products are INT16 and accumulated into INT32.

2) FLOP count

Each output element performs (K=576) MACs.

$$\text{Output elements} = 52 \cdot 52 \cdot 128 = 346112$$

$$\text{MACs} = 346112 \cdot 576 = 199360512$$

$$\text{FLOPs} = 2 \cdot 199360512 = 398721024$$

3) Off-chip bytes

The weight matrix is reused across all output spatial positions.

No reuse

Each MAC reads one activation byte and one weight byte. Each output is written once.

$$\text{Activation reads} = 199360512 \cdot 1 = 199360512 \text{ B}$$

$$\text{Weight reads} = 199360512 \cdot 1 = 199360512 \text{ B}$$

$$\text{Output writes} = 52 \cdot 52 \cdot 128 \cdot 1 = 346112 \text{ B}$$

$$\text{Total bytes}_{low} = 199360512 + 199360512 + 346112 = 399067136 \text{ B}$$

Perfect on-chip weight reuse

Weights are loaded once and reused on chip. The input and output feature maps each cross the off-chip boundary once.

$$\text{Input} = 52 \cdot 52 \cdot 64 = 173056 \text{ B}$$

$$\text{Weights} = 3 \cdot 3 \cdot 64 \cdot 128 = 73728 \text{ B}$$

$$\text{Output} = 52 \cdot 52 \cdot 128 = 346112 \text{ B}$$

$$\text{Total bytes}_{high} = 173056 + 73728 + 346112 = 592896 \text{ B}$$

4) Arithmetic intensity

$$AI = \frac{\text{FLOPs}}{\text{Bytes}}$$

No reuse:

$$AI_{low} = \frac{398721024}{399067136} = 0.999 \approx 1.0 \text{ FLOP/B}$$

Perfect weight reuse:

$$AI_{high} = \frac{398721024}{592896} = 672.5 \text{ FLOP/B}$$

At the real operating point, the kernel is near the high-AI bound because weights are buffered and reused on chip.

5) Bottleneck and highest-leverage change

The design as it stands is compute-bound. With weight reuse, the arithmetic intensity is about 672.5 FLOP/B, so the kernel is to the right of the bandwidth ridge and therefore limited by MAC throughput.

The highest-leverage change is therefore to increase the MAC throughput, after fixing the timing path through tap selection, multiply, and accumulation.